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Exercises

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Course Name : Fundamentals of Data Science

1. Find mean, median, mode and range for: 12, 15, 18, 15, 20, 22, 15, 25.

Given Data

12, 15, 18, 15, 20, 22, 15, 25

Step 1: Arrange the Data in Ascending Order

First, arrange the given observations from smallest to largest.

12, 15, 15, 15, 18, 20, 22, 25

Number of observations, $n = 8$

Step 2: Calculate Mean

The mean is the average of all the observations. It is calculated by dividing the sum of all observations by the total number of observations.

Formula:

$$\text{Mean} = \text{Sum of all observations} / \text{Number of observations}$$

Sum of observations:

$$12 + 15 + 18 + 15 + 20 + 22 + 15 + 25 = 142$$

Therefore:

$$\text{Mean} = 142 / 8$$

$$\text{Mean} = 17.75$$

Therefore, the mean of the given dataset is 17.75.

Step 3: Calculate Median

The median is the middle value of an ordered dataset.

The arranged data is:

12, 15, 15, 15, 18, 20, 22, 25

Since there are 8 observations, the number of observations is even. Therefore, the median is calculated by taking the average of the two middle values.

The 4th observation is 15.

The 5th observation is 18.

Formula:

$$\text{Median} = (4\text{th value} + 5\text{th value}) / 2$$

$$\text{Median} = (15 + 18) / 2$$

$$\text{Median} = 33 / 2$$

$$\text{Median} = 16.5$$

Therefore, the median is 16.5.

Step 4: Calculate Mode

The mode is the value that occurs most frequently in the dataset.

In the given data, the value 15 occurs three times.

The other values occur only once.

Therefore:

$$\text{Mode} = 15$$

Hence, the mode is 15.

Step 5: Calculate Range

Range represents the difference between the largest and smallest observations.

Formula:

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

$$\text{Maximum Value} = 25$$

$$\text{Minimum value} = 12$$

$$\text{Range} = 25 - 12$$

$$\text{Range} = 13$$

Therefore, the range is 13.

Final Answer

$$\text{Mean} = 17.75$$

$$\text{Median} = 16.5$$

Mode = 15

Range = 13

Interpretation

The mean of the dataset is 17.75, which represents the average value of all observations. The median is 16.5, which represents the central position of the ordered dataset. The mode is 15 because it occurs more frequently than the other values. The range is 13, indicating that the difference between the highest and lowest observations is 13.

2. Construct a frequency distribution for: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.

Given Data

2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4

A frequency distribution shows how many times each individual value occurs in the dataset.

Step 1: Arrange the Data

Arrange the observations in ascending order.

2, 2, 2, 3, 3, 3, 4, 4, 4, 5, 5, 6

There are 12 observations in total.

Step 2: Identify the Different Values

The different values present in the dataset are:

2, 3, 4, 5 and 6.

Step 3: Count the Frequency of Each Value

The value 2 occurs 3 times.

The value 3 occurs 3 times.

The value 4 occurs 3 times.

The value 5 occurs 2 times.

The value 6 occurs 1 time.

Step 4: Construct the Frequency Distribution

Value	Frequency
2	3
3	3
4	3

Value	Frequency
5	2
6	1
Total	12

Step 5: Verify the Frequency

The sum of all frequencies should be equal to the total number of observations.

Total frequency = $3 + 3 + 3 + 2 + 1$

Total frequency = 12

Since the original dataset also contains 12 observations, the frequency distribution is correct.

Interpretation

The values 2, 3 and 4 have the highest frequency because each occurs three times. The value 5 occurs twice, while the value 6 occurs only once. Therefore, 2, 3 and 4 are the most frequently occurring values in the dataset.

3. Find Q1, Q2, Q3 and IQR for: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.

Given Data

5, 8, 10, 12, 15, 18, 20, 22, 25, 30

Quartiles divide an ordered dataset into four equal parts. Q1 represents the first quartile, Q2 represents the second quartile or median, and Q3 represents the third quartile.

Step 1: Arrange the Data

The data is already arranged in ascending order.

5, 8, 10, 12, 15, 18, 20, 22, 25, 30

Number of observations:

$n = 10$

Step 2: Find Q2

Q2 is the second quartile and is also called the median.

Since there are 10 observations, the median is the average of the 5th and 6th observations.

5th observation = 15

6th observation = 18

Formula:

$$Q2 = ((n/2)\text{th value} + (n+1)/2\text{th value}) / 2$$

$$Q2 = (15 + 18) / 2$$

$$Q2 = 33 / 2$$

$$Q2 = 16.$$

Therefore:

$$Q2 = 16.5$$

Step 3: Find Q1

Q1 is the median of the lower half of the dataset.

Lower half:

5, 8, 10, 12, 15

The middle value of this group is 10.

Therefore:

$$Q1 = 10$$

Step 4: Find Q3

Q3 is the median of the upper half of the dataset.

Upper half:

18, 20, 22, 25, 30

The middle value of this group is 22.

Therefore:

$$Q3 = 22$$

Step 5: Calculate IQR

IQR stands for Interquartile Range. It represents the spread of the middle 50% of the observations.

Formula:

$IQR = Q3 - Q1$

$$IQR = 22 - 10$$

$$IQR = 12$$

Therefore:

$$IQR = 12$$

Final Answer

$$Q1 = 10$$

$$Q2 = 16.5$$

$$Q3 = 22$$

$$IQR = 12$$

Interpretation

Q1 is 10, which represents the first quartile of the dataset. Q2 is 16.5 and represents the median. Q3 is 22, representing the third quartile. The IQR is 12, which indicates that the middle 50% of the observations are spread across a range of 12 units.

4. Detect outliers using the IQR method for: 10, 12, 13, 15, 16, 17, 18, 20, 45.

Given Data

10, 12, 13, 15, 16, 17, 18, 20, 45

The IQR method is commonly used to identify observations that are unusually high or unusually low.

Step 1: Arrange the Data

The data is already arranged in ascending order.

10, 12, 13, 15, 16, 17, 18, 20, 45

There are 9 observations.

The middle observation is 16.

Step 2: Find Q1

The lower half of the dataset is:

10, 12, 13, 15

Since there are four values, Q1 is the average of the two middle values.

$$Q1 = (12 + 13) / 2$$

$$Q1 = 25 / 2$$

$$Q1 = 12.5$$

Therefore:

$$Q1 = 12.5$$

Step 3: Find Q3

The upper half of the dataset is:

17, 18, 20, 45

Q3 is the average of the two middle values.

$$Q3 = (18 + 20) / 2$$

$$Q3 = 38 / 2$$

$$Q3 = 19$$

Therefore:

$$Q3 = 19$$

Step 4: Calculate IQR

Formula:

$$IQR = Q3 - Q1$$

$$IQR = 19 - 12.5$$

$$IQR = 6.5$$

Therefore:

$$IQR = 6.5$$

Step 5: Calculate the Lower Fence

Formula:

$$\text{Lower Fence} = Q1 - 1.5 \times IQR$$

$$\text{Lower Fence} = 12.5 - 1.5 \times 6.5$$

$$\text{Lower Fence} = 12.5 - 9.75$$

$$\text{Lower Fence} = 2.75$$

Step 6: Calculate the Upper Fence

Formula:

$$\text{Upper Fence} = Q3 + 1.5 \times IQR$$

$$\text{Upper Fence} = 19 + 1.5 \times 6.5$$

$$\text{Upper Fence} = 19 + 9.75$$

$$\text{Upper Fence} = 28.75$$

Therefore, the normal range according to the IQR method is from 2.75 to 28.75.

Step 7: Identify the Outlier

The largest value in the dataset is 45.

Since:

$$45 > 28.75$$

the value 45 lies above the upper fence.

Therefore:

Outlier = 45

Final Answer

$Q1 = 12.5$

$Q3 = 19$

$IQR = 6.5$

Lower Fence = 2.75

Upper Fence = 28.75

Outlier = 45

Interpretation

The value 45 is significantly larger than the other observations. Since it lies outside the upper fence of 28.75, it is identified as an outlier using the IQR method.

5.Determine the skewness (Asymmetry) of: 5, 6, 7, 7, 8, 8, 9, 20.

Given Data

5, 6, 7, 7, 8, 8, 9, 20

Skewness describes the degree of asymmetry in a dataset. A distribution can be approximately symmetric, positively skewed or negatively skewed.

Step 1: Calculate the Mean

First, calculate the sum of all observations.

$$5 + 6 + 7 + 7 + 8 + 8 + 9 + 20 = 70$$

Number of observations:

$$n = 8$$

Formula:

$$\text{Mean} = \text{Sum of observations} / \text{Number of observations}$$

$$\text{Mean} = 70/8$$

$$\text{Mean} = 8.75$$

Therefore:

$$\text{Mean} = 8.75$$

Step 2: Calculate the Median

The data is already arranged:

5, 6, 7, 7, 8, 8, 9, 20

There are 8 observations, so the median is the average of the 4th and 5th observations.

4th observation = 7

5th observation = 8

Median = $(7 + 8) / 2$

Median = $15 / 2$

Median = 7.5

Therefore:

Median = 7.5

Step 3: Compare Mean and Median

Mean = 8.75

Median = 7.5

Therefore:

Mean > Median

When the mean is greater than the median, it generally indicates positive skewness.

Step 4: Identify the Direction of Skewness

Most of the observations are between 5 and 9.

However, the value 20 is much larger than the remaining observations.

This large value pulls the mean toward the right side and creates a longer tail on the right side of the distribution.

Therefore, the distribution is positively skewed or right-skewed.

Final Answer

The dataset is positively skewed or right-skewed.

Interpretation

The observation 20 is responsible for much of the asymmetry in the dataset. It is considerably larger than the other observations and increases the mean to 8.75, while the median remains at 7.5. This difference between the mean and median indicates positive skewness.

6.A uniformly distributed variable ranges from 10 to 50. Find $P(20 < X < 35)$.

Given

A uniformly distributed variable ranges from 10 to 50.

We need to find:

$$P(20 < X < 35)$$

Step 1: Understand the Uniform Distribution

In a continuous uniform distribution, every value within the given range has an equal probability of occurring.

For a uniform distribution, the probability between two values can be calculated using:

$$P(a < X < b) = (b - a) / (\text{Upper limit} - \text{Lower limit})$$

Step 2: Calculate the Total Range

$$\text{Lower limit} = 10$$

$$\text{Upper limit} = 50$$

$$\text{Total range} = 50 - 10$$

$$\text{Total range} = 40$$

Step 3: Calculate the Required Interval

The required interval is from 20 to 35.

$$\text{Required interval} = 35 - 20$$

$$\text{Required interval} = 15$$

Step 4: Calculate the Probability

$$P(20 < X < 35) = 15 / 40$$

$$P(20 < X < 35) = 0.375$$

Step 5: Convert the Probability into Percentage

$$\text{Probability} = 0.375 \times 100$$

$$\text{Probability} = 37.5\%$$

Final Answer

$$P(20 < X < 35) = 0.375$$

Therefore, the probability is 37.5%.

Interpretation

There is a 37.5% probability that a randomly selected value from this uniform distribution will fall between 20 and 35.

7.A normal distribution has mean 100 and standard deviation 15. Find the Z-score for X=130.

Given

Mean = 100

Standard deviation = 15

$X = 130$

A Z-score tells us how many standard deviations an observation is away from the mean.

Step 1: Write the Z-score Formula

$$Z = (X - \text{Mean}) / \text{Standard deviation}$$

Step 2: Substitute the Values

$$Z = (130 - 100) / 15$$

Step 3: Perform the Calculation

$$Z = 30 / 15$$

$$Z = 2$$

Final Answer

$$Z = 2$$

Interpretation

The Z-score is 2. This means that the value 130 is two standard deviations above the mean of 100.

Since the Z-score is positive, the value 130 lies above the mean.

8. For the same normal distribution, find $P(X < 130)$.**Given**

Mean = 100

Standard deviation = 15

$X = 130$

We need to find:

$$P(X < 130)$$

Step 1: Calculate the Z-score

The Z-score formula is:

$$Z = (X - \text{Mean}) / \text{Standard deviation}$$

Substituting the given values:

$$Z = (130 - 100) / 15$$

$$Z = 30 / 15$$

$$Z = 2$$

Therefore:

$$Z = 2.00$$

Step 2: Use the Standard Normal Distribution Table

For a Z-score of 2.00, the cumulative probability from the standard normal distribution table is:

$$P(Z < 2.00) = 0.9772$$

Therefore:

$$P(X < 130) = 0.9772$$

Step 3: Convert into Percentage

$$\text{Probability} = 0.9772 \times 100$$

$$\text{Probability} = 97.72\%$$

Final Answer

$$P(X < 130) = 0.9772$$

or

$$P(X < 130) = 97.72\%$$

Interpretation

There is approximately a 97.72% probability that a randomly selected observation from this normal distribution will be less than 130.

The probability of an observation being greater than 130 is:

$$P(X > 130) = 1 - 0.9772$$

$$P(X > 130) = 0.0228$$

Therefore, approximately 2.28% of observations are expected to be greater than 130.

9. Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.

Example Dataset

Consider the following dataset:

5, 6, 7, 8, 9

We will compare the mean and median before introducing an outlier, after introducing an outlier and after treating the outlier.

Step 1: Calculate the Original Mean

Formula:

$$\text{Mean} = \text{Sum of observations} / \text{Number of observations}$$

$$\text{Mean} = (5 + 6 + 7 + 8 + 9) / 5$$

$$\text{Mean} = 35 / 5$$

$$\text{Mean} = 7$$

Therefore:

$$\text{Mean} = 7$$

Step 2: Calculate the Original Median

The data is already arranged:

5, 6, 7, 8, 9

There are five observations, so the middle observation is 7.

Therefore:

$$\text{Median} = 7$$

Before the outlier:

$$\text{Mean} = 7$$

$$\text{Median} = 7$$

Step 3: Introduce an Extreme Value

Suppose an extreme value of 50 is added to the dataset.

New dataset:

5, 6, 7, 8, 9, 50

The value 50 is considerably larger than all the other observations and can be treated as an extreme value.

Step 4: Calculate the Mean After Adding the Outlier

$$\text{Mean} = (5 + 6 + 7 + 8 + 9 + 50) / 6$$

$$\text{Mean} = 85 / 6$$

$$\text{Mean} = 14.17$$

Therefore:

$$\text{Mean} = 14.17$$

The mean has increased from 7 to 14.17 because of the extreme value.

Step 5: Calculate the Median After Adding the Outlier

There are now six observations.

The two middle observations are 7 and 8.

$$\text{Median} = (7 + 8) / 2$$

$$\text{Median} = 15 / 2$$

$$\text{Median} = 7.5$$

Therefore:

$$\text{Median} = 7.5$$

The median has changed only slightly from 7 to 7.5.

Step 6: Treat the Outlier

Suppose the value 50 is identified as an outlier and removed.

The dataset becomes:

5, 6, 7, 8, 9

$$\text{Mean} = 7$$

$$\text{Median} = 7$$

Step 7: Compare the Results

Before outlier:

$$\text{Mean} = 7$$

$$\text{Median} = 7$$

With outlier:

$$\text{Mean} = 14.17$$

$$\text{Median} = 7.5$$

After outlier treatment:

$$\text{Mean} = 7$$

$$\text{Median} = 7$$

Interpretation

The mean is highly sensitive to extreme values because every observation is included in its calculation. When the value 50 is added, the mean increases significantly from 7 to 14.17.

The median is much less affected by the outlier. It changes only from 7 to 7.5.

After removing the outlier, both the mean and median return to their original values.

Therefore, the median is considered a more robust measure of central tendency when extreme values are present in a dataset.

10. Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers and skewness.

Student-Mark Dataset

Consider the following marks obtained by 10 students:

45, 50, 52, 55, 60, 62, 65, 68, 70, 95

Exploratory Data Analysis, commonly known as EDA, is the process of examining and summarizing a dataset to understand its important characteristics. It involves studying central tendency, dispersion, distribution, quartiles, outliers and skewness.

Step 1: Dataset Overview

The dataset contains the marks of 10 students.

Student 1 = 45

Student 2 = 50

Student 3 = 52

Student 4 = 55

Student 5 = 60

Student 6 = 62

Student 7 = 65

Student 8 = 68

Student 9 = 70

Student 10 = 95

Number of observations:

$n = 10$

Minimum mark = 45

Maximum mark = 95

Step 2: Calculate Mean

The mean represents the average mark obtained by the students.

Formula:

Mean = Sum of all observations / Number of observations

Sum of marks:

$$45 + 50 + 52 + 55 + 60 + 62 + 65 + 68 + 70 + 95 = 622$$

Therefore:

$$\text{Mean} = 622 / 10$$

$$\text{Mean} = 62.2$$

Therefore, the average mark of the students is 62.2.

Step 3: Calculate Median

The data is already arranged in ascending order:

45, 50, 52, 55, 60, 62, 65, 68, 70, 95

There are 10 observations, so the median is calculated using the 5th and 6th observations.

$$\text{5th observation} = 60$$

$$\text{6th observation} = 62$$

$$\text{Median} = (60 + 62) / 2$$

$$\text{Median} = 122 / 2$$

$$\text{Median} = 61$$

Therefore:

$$\text{Median} = 61$$

Step 4: Calculate Mode

The mode is the value that occurs most frequently.

In this dataset, every mark occurs only once.

Therefore:

Mode = No mode

Step 5: Calculate Range

The range represents the difference between the maximum and minimum observations.

Formula:

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

$$\text{Range} = 95 - 45$$

$$\text{Range} = 50$$

Therefore:

$$\text{Range} = 50$$

This means that the difference between the highest and lowest marks is 50.

Step 6: Calculate Q1

The lower half of the dataset is:

45, 50, 52, 55, 60

The middle value is 52.

Therefore:

$$Q1 = 52$$

Step 7: Calculate Q2

Q2 is the median of the complete dataset.

Therefore:

$$Q2 = 61$$

Step 8: Calculate Q3

The upper half of the dataset is:

62, 65, 68, 70, 95

The middle value is 68.

Therefore:

$$Q3 = 68$$

Step 9: Calculate IQR

The Interquartile Range represents the spread of the middle 50% of the observations.

Formula:

$$IQR = Q3 - Q1$$

$$IQR = 68 - 52$$

$$IQR = 16$$

Therefore:

$$IQR = 16$$

Step 10: Detect Outliers Using the IQR Method

The IQR method identifies observations that fall significantly below or above the central range.

First, calculate the lower fence.

Formula:

$$\text{Lower Fence} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Lower Fence} = 52 - 1.5 \times 16$$

$$\text{Lower Fence} = 52 - 24$$

$$\text{Lower Fence} = 28$$

Now calculate the upper fence.

Formula:

$$\text{Upper Fence} = Q3 + 1.5 \times \text{IQR}$$

$$\text{Upper Fence} = 68 + 1.5 \times 16$$

$$\text{Upper Fence} = 68 + 24$$

$$\text{Upper Fence} = 92$$

Therefore, the expected range of non-outlier observations is from 28 to 92.

The dataset contains the value 95.

Since:

$$95 > 92$$

the value 95 lies outside the upper fence.

Therefore:

$$\text{Outlier} = 95$$

Step 11: Analyze the Distribution

Most of the student marks are concentrated between 45 and 70.

However, the value 95 is considerably higher than the remaining observations.

This creates a longer tail toward the right side of the distribution.

Therefore, the distribution is right-skewed.

Step 12: Analyze Skewness

$$\text{Mean} = 62.2$$

$$\text{Median} = 61$$

Since:

$$\text{Mean} > \text{Median}$$

the dataset indicates positive skewness.

The high observation of 95 pulls the mean upward and contributes to the right-skewed distribution.

Therefore:

Skewness = Positive or Right-skewed

Step 13: Analyze Central Tendency

The mean of 62.2 represents the average performance of the students.

The median of 61 represents the middle position of the ordered dataset.

Because the dataset contains the extreme value 95, the mean is affected by this observation. The median is less sensitive to extreme values.

Therefore, the median can be considered a more representative measure of typical student performance when the outlier is present.

Step 14: Analyze Dispersion

The range is 50, indicating that there is a considerable difference between the lowest and highest marks.

However, the range is strongly affected by the extreme value 95.

The IQR is 16 and represents the spread of the middle 50% of the observations. Since the IQR is based on Q1 and Q3, it is less affected by extreme observations.

Therefore, the IQR provides a more reliable description of the central spread of the dataset.

Step 15: Analyze the Overall Distribution

The majority of students have marks between 45 and 70. The marks gradually increase across the dataset, while the value 95 is considerably separated from the other observations.

This indicates that the dataset is not perfectly symmetric.

The high value of 95 creates a longer right tail, making the distribution positively skewed.

Step 16: Overall EDA Summary

Number of students = 10

Minimum mark = 45

Maximum mark = 95

Mean = 62.2

Median = 61

Mode = No mode

Range = 50

Q1 = 52

Q2 = 61

Q3 = 68

IQR = 16

Lower Fence = 28

Upper Fence = 92

Outlier = 95

Distribution = Right-skewed

Skewness = Positive